Tut 6.

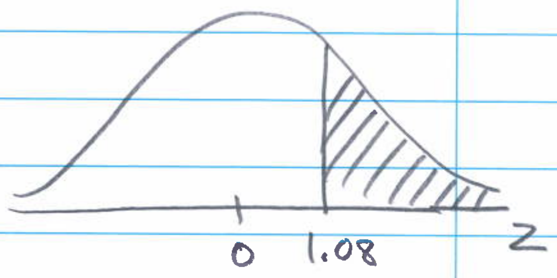
② Standard normal distribution

$$Z \sim N(0,1)$$

(a) $P(Z > 1.08)$

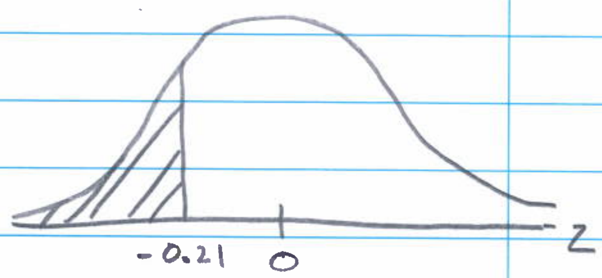
$= 1 - P(Z < 1.08)$

$= 1 - 0.8599 = \underline{0.1401}$



(b) $P(Z < -0.21)$

$= \underline{0.4168}$

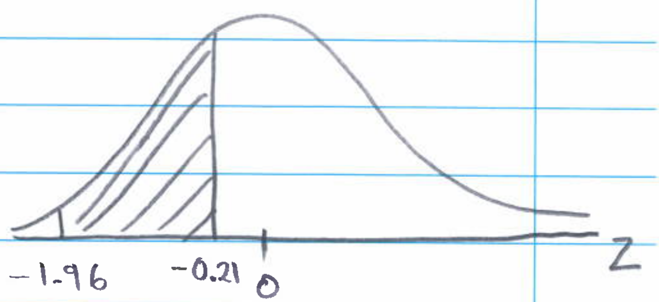


(c) $P(-1.96 < Z < -0.21)$

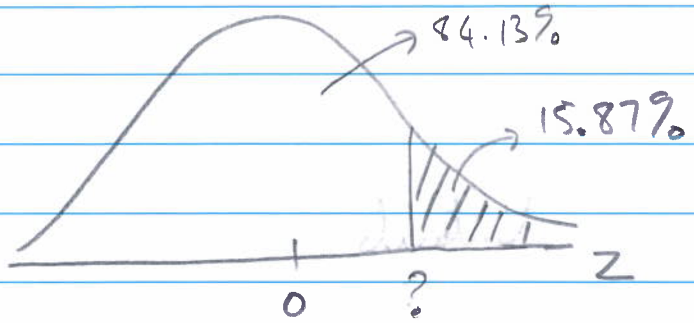
$= P(Z < -0.21) - P(Z < -1.96)$

$= 0.4168 - 0.025$

$= \underline{0.3918}$



(d)

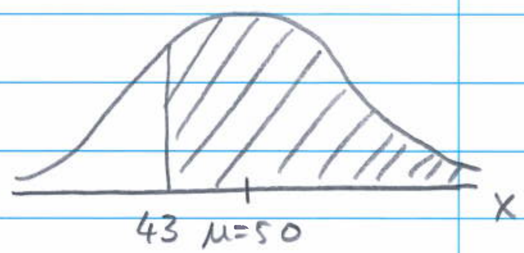


Thus, $P(Z < ?) = 0.8413$ or $P(Z > ?) = 0.1587$

$\therefore\ ? = 1$

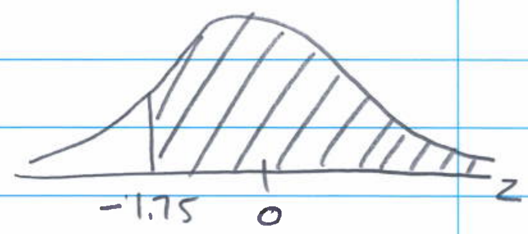
3. Given $X \sim N(\mu = 50, \sigma^2 = 4^2)$

(a) $P(X > 43)$



$= P\left(\frac{X-\mu}{\sigma} > \frac{43-50}{4}\right)$

$= P(Z > -1.75)$



$= 1 - P(Z < -1.75)$

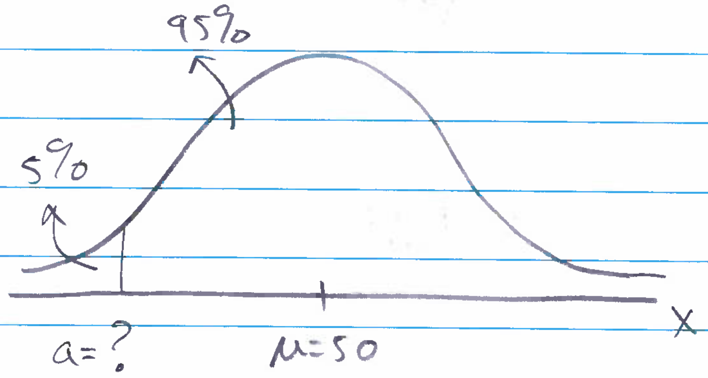
$= 1 - 0.0401 = \underline{0.9599}$

(b) $P(X < 42) = P\left(\frac{X-\mu}{\sigma} < \frac{42-50}{4}\right)$

$= P(Z < -2)$

$= \underline{0.0228}$

(c)



$P(X < a) = 0.05$

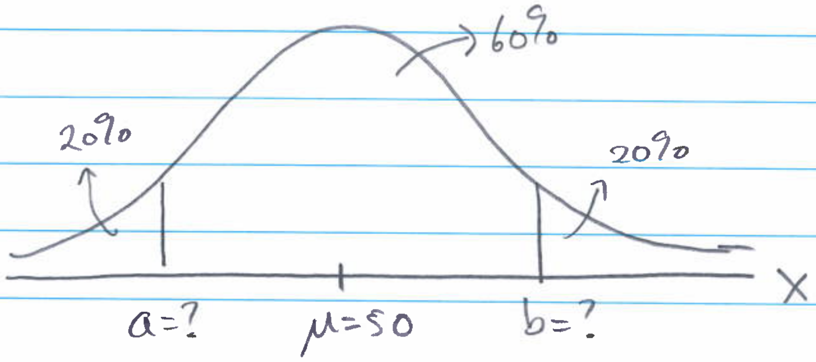
Use $\quad a = \mu + Z\sigma$

$P_5 = a = \mu + Z_5\,\sigma \qquad$ (P_5 = 5th Percentile)

$= 50 + (-1.645)\,4$

$= \underline{43.42}$

(d)



$P_{20} = a = \mu + Z_{20}\,\sigma = 50 + (-0.84)\,4 = \underline{46.64}$

$P_{80} = b = \mu + Z_{80}\,\sigma = 50 + (0.84)\,4 = \underline{53.36}$

4. X = final mark

$X \sim N(\mu = 73, \sigma^2 = 8^2)$

(a) $P(X < 91)$

$= P\left(\frac{X-\mu}{\sigma} < \frac{91-73}{8}\right)$

$= P(Z < 2.25) = 0.9878$

(b) $P(65 < X < 89)$

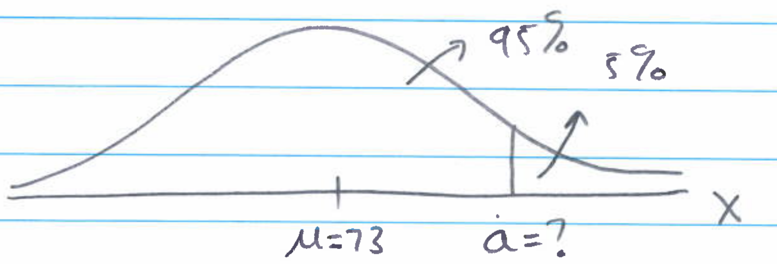
$= P(X < 89) - P(X < 65)$

$= P\left(\frac{X-\mu}{\sigma} < \frac{89-73}{8}\right) - P\left(\frac{X-\mu}{\sigma} < \frac{65-73}{8}\right)$

$= P(Z < 2) - P(Z < -1)$

$= 0.9772 - 0.1587 = 0.8185$

(c)



Use $a = \mu + Z\sigma$

$P_{95} = a = \mu + Z_{95}\sigma = 73 + (1.645)8 = 86.16$

(d)

$X \sim N(\mu = 73, \sigma^2 = 8^2)$ $\Rightarrow$ Course 1

$Y \sim N(\mu = 62, \sigma^2 = 3^2)$ $\Rightarrow$ Course 2

Course 1

$P(X > a) = 0.1$

$P_{90} = a = \mu + Z_{90}\sigma$

$= 73 + (1.28)8$

$= 83.24$

Course 2

$P(Y > b) = 0.1$

$P_{90} = b = \mu + Z_{90}\sigma$

$= 62 + (1.28)3$

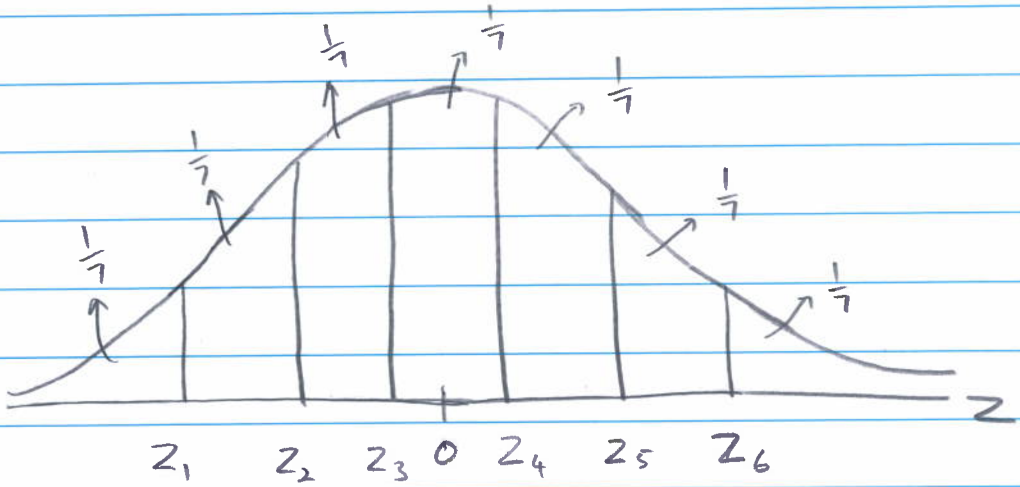
$= 65.84$

For course 2 you need a lower mark to get a distinction

Thus, if you get 81 for course 1, you do not get a distinction

If you get 68 for course 2, you get a distinction.

6



$Z \sim N(\mu=0, \sigma^2=1)$

Use $a = \mu + z\sigma$ or

$P(Z < z_1) = 0.1429$, thus $z_1 = -1.07$

$P(Z < z_2) = 0.2857$, thus $z_2 = -0.57$

$P(Z < z_3) = 0.4286$, thus $z_3 = -0.18$

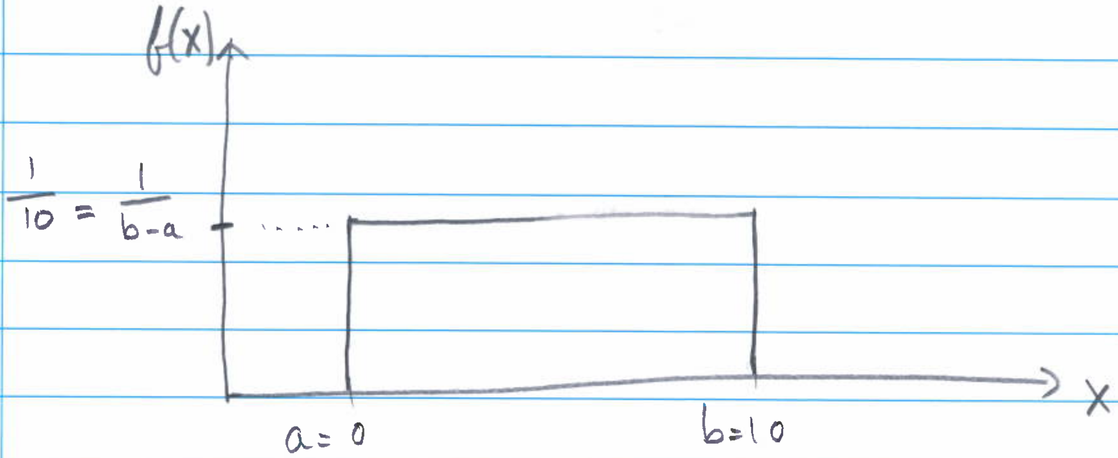
$P(Z < z_4) = 0.5714$, thus $z_4 = 0.18$

$P(Z < z_5) = 0.7143$, thus $z_5 = 0.57$

$P(Z < z_6) = 0.8571$, thus $z_6 = 1.07$

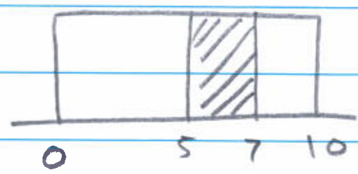
(8)

$$X \sim \text{Uniform}(a=0,\ b=10)$$



(a) $P(5 < X < 7) = (7-5) \times \frac{1}{10}$

$= 0.4$



(b) $P(2 < X < 3) = (3-2) \times \frac{1}{10}$

$= 0.1$

(c) $\mu = E(X) = \text{average} = \frac{a+b}{2} = 5$

(d) $\sigma = \sqrt{Var(X)} = \text{standard deviation}$

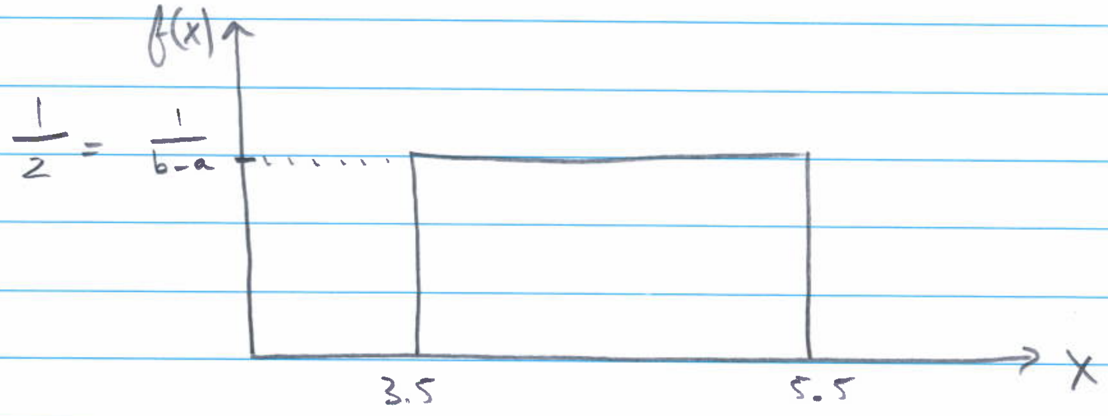
$$= \sqrt{\frac{(b-a)^2}{12}}$$

$$= \sqrt{8.3333} = 2.8868$$

9

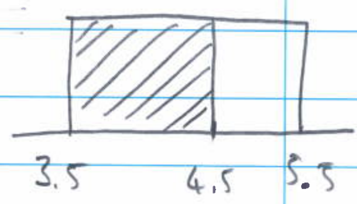
$X =$ download time

$X \sim \text{Uniform}(a = 3.5, b = 5.5)$



(a) $P(X < 4.5) = (4.5 - 3.5) \times \frac{1}{2}$

$= 0.5$



(b) $P(X > 4) = (5.5 - 4) \times \frac{1}{2}$

$= 0.75$

(c) $P(4 < X < 4.5) = (4.5 - 4) \times \frac{1}{2}$

$= 0.25$

(d) $\mu = \frac{b+a}{2} = 4.5$ minutes

$\sigma = \sqrt{\frac{(b-a)^2}{12}} = 0.5774$ minutes

(ii)

X = time

$X \sim \text{Exponential}(\lambda = 5)$

$P(X < t) = 1 - e^{-\lambda t}$

(a) $P(X < 0.3) = 1 - e^{-5(0.3)}$

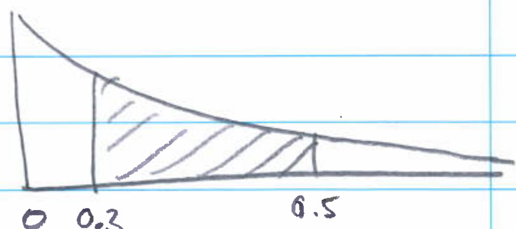
$= 1 - 0.22313 = \underline{0.7769}$

(b) $P(X > 0.3) = 1 - \{1 - e^{-\lambda t}\}$

$= e^{-\lambda t} = e^{-5(0.3)} = \underline{0.2231}$

(c) $P(X > 0.3 \text{ and } X < 0.5)$

$= P(0.3 < X < 0.5)$



$= P(X < 0.5) - P(X < 0.3)$

$= 1 - e^{-5(0.5)} - \{1 - e^{-5(0.3)}\}$

$= e^{-5(0.3)} - e^{-5(0.5)}$

$= 0.2231 - 0.0821 = \underline{0.141}$

(d) $P(X < 0.3 \text{ or } X > 0.5) = P(X < 0.3) + P(X > 0.5)$

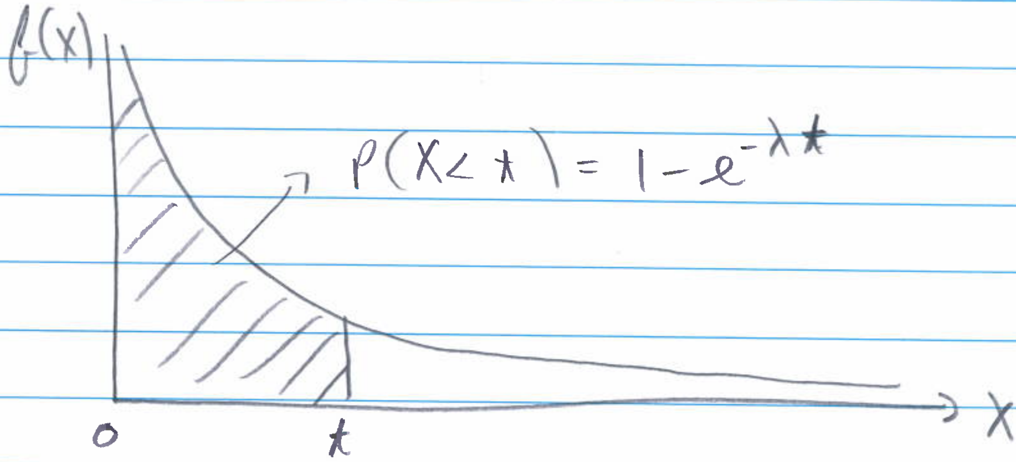
$= 0.7769 + 0.0821 = \underline{0.859}$

12

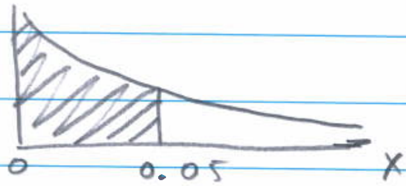
X = arrival time

$X \sim \text{Exponential}(\lambda = 50)$

Area of opportunity = minute



(a) $P(X < 0.05)$



$= 1 - e^{-(50)(0.05)}$

$= 1 - 0.082085 = \underline{0.9179}$

(b) $P(X < \frac{1}{60}) = 1 - e^{-50(\frac{1}{60})}$

$= 1 - 0.4346 = \underline{0.5654}$

(c) If $X \sim \text{Exponential}(\lambda = 60)$

→ (a) $P(X < 0.05) = 1 - e^{-60(0.05)}$

$= 1 - 0.049787 = \underrightarrow{0.9502}$

(b) $P(X < \frac{1}{60}) = 1 - e^{-60(\frac{1}{60})}$

$= 1 - 0.367879 = \underrightarrow{0.6321}$

(d) If $X \sim \text{Exponential}(\lambda = 30)$

→ (a) $P(X < 0.05) = 1 - e^{-30(0.05)}$

$= \underrightarrow{0.7769}$

→ (b) $P(X < \frac{1}{60}) = 1 - e^{-30(\frac{1}{60})}$

$= \underrightarrow{0.3935}$

(e) Extra: $\mu = \frac{1}{\lambda}$

$\sigma^2 = \frac{1}{\lambda^2}$

$\sigma = \frac{1}{\lambda}$

(17)

X = number of passengers who eat pretzels

Y = number of passengers who do not eat pretzels

Thus,

$$X \sim \text{Binomial}\left(n=150, \pi = \frac{1}{3}\right)$$

$$Y \sim \text{Binomial}\left(n=150, \pi = \frac{2}{3}\right)$$

(a) Use $X_a \sim N(n\pi, n\pi(1-\pi))$

$$\mu = n\pi = 50, \quad \sigma = \sqrt{n\pi(1-\pi)} = 5.7735$$

$$P(X=60) \approx P(59.5 < X_a < 60.5)$$

$$= P(X_a < 60.5) - P(X_a < 59.5)$$

$$= P\left(\frac{X_a-\mu}{\sigma} < \frac{60.5-50}{5.7735}\right) - P\left(\frac{X_a-\mu}{\sigma} < \frac{59.5-50}{5.7735}\right)$$

$$= P(Z < 1.82) - P(Z < 1.65)$$

$$= 0.9656 - 0.9505$$

$$= \underline{0.0151}$$

(b) $P(X \geq 60) \approx P(X_a > 59.5)$

$$= P\left(\frac{X_a - \mu}{\sigma} > \frac{59.5 - 50}{5.7735}\right)$$

$$= P(Z > 1.65)$$

$$= 1 - P(Z < 1.65)$$

$$= 1 - 0.9505 = \underline{0.0495}$$

(c) $P(X < 60) = P(X \leq 59)$

$$\approx P(X_a < 59.5)$$

$$= P\left(\frac{X_a - \mu}{\sigma} < \frac{59.5 - 50}{5.7735}\right)$$

$$= P(Z < 1.65)$$

$$= \underline{0.9505}$$

(d) Thus, 70 passengers can get pretzels

$P(X \leq 70)$ is the probability that 70 or less gets pretzels

$1 - P(X \leq 70) = P(X > 70)$ is thus the probability that a passenger does not get pretzels.

$$P(X > 70) = P(X \geq 71)$$

$$\approx P(X_a > 70.5)$$

$$= P\left(\frac{X_a - \mu}{\sigma} > \frac{70.5 - 50}{5.7735}\right)$$

$$= P(Z > 3.55)$$

$$= 1 - P(Z < 3.55)$$

$$= 1 - 0.99981$$

$$= \underline{0.00019}$$